Chandramauli Chakraborty

Curriculum Vitae

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Education

2024–present **PhD in Statistics**, *University of Chicago*
Advisor - Prof. Cong Ma

2022–2024 **Master of Statistics (M.Stat)**, *Indian Statistical Institute (I.S.I.), Kolkata, India*
- First division with Distinction.

2019–2022 **Bachelor of Statistics (B.Stat)**, *Indian Statistical Institute (I.S.I.), Kolkata, India*
- Scored 82% in aggregate, First division with Distinction.

2017–2019 **Higher Secondary Examination (Standard 12)**, *Bhavan's Gangabux Kanoria Vidyamandir, Kolkata, West Bengal*
Scored 93.8% in aggregate, with 95% in Mathematics, Physics and Chemistry

2007–2017 **Secondary Examination (Standard 10)**, *Don Bosco H.S. School, Silchar, Assam*
Scored 10 CGPA in aggregate, with A1 grade (above 95%) in Mathematics and Science

Academic achievements

2024 Received the prestigious **Neubauer Fellowship** from the University of Chicago.

2024 Got an invitation for the prestigious **Sabyasachi Roy Memorial Gold Medal**.

2023 Got funded by the ACM India-IARCS travel grants

2022 Got an invitation for the prestigious **D. Basu Gold Medal Award**.

2019–2024 Received Prize Money and Stipends each semester from the Indian Statstical Institute, Kolkata.

2019 Qualified **IISER Aptitude Test**, was offered a seat at IISER.

2019 **AIR 86** Banaras Hindu University Entrance Examination.

2019 Qualified JEE Mains and JEE Advanced Examination, was among the **top 2%**.

2019 Scored 790/800 in Mathematics section of the SAT.

2018 Qualified the written exam of **Kishore Vaigyanik Protsahan Yojana (KVPY)**.

2017 Qualified the **National Talent Search Examination (NTSE)** State-Level.

2016 **Regional Mathematics Olympiad (RMO)** awardee. Appeared for **Indian National Mathematics Olympiad (INMO)**.

Summer Projects/ Internships

June 2023 – August 2023 Worked with **Prof. Larry Wasserman** and **Prof. Arun Kumar Kuchibhotla** in several topics of generative models, and particularly working with theoretical aspects of the rectified flow.

June 2022 – August 2022 Participated in **Big Data Summer Institute** (BDSI) conducted by the Department of Biostatistics funded by NIH, University of Michigan, where I have worked in a *Data Mining* group under the guidance of Prof. Johann Gagnon-Bartsch where I received advanced statistical and computational training through project and lectures. I have used statistical spatial epidemiology techniques in Python to forecast numbers of "deficit" STI cases in each county of the US due to decreases in surveillance testing during the pandemic. We attended lectures included machine learning, Bayesian statistics, deep learning, natural language processing, missing data analysis, model selection, unsupervised learning, and more.

June 2021 – August 2021: Worked in the project titled "Selecting balls from urns", as a part of the *Polymath Jr. program*, an REU program conducted by the Williams College, partially supported by NSF, under the supervision of Enrique Treviño, where we were introduced to various models of selecting balls from an urn, found some interesting properties of a particular method of selecting balls from the urn and proved few conjectures as posed by the paper To Replace or Not to Replace — That is the Question.

January 2021–January 2022: Worked on *Modelling COVID-19 pandemic and optimization of economy and period of lockdown* under the supervision of Prof. Gopal K. Basak.

January 2021–June 2021: Worked on a *Machine Learning* project under the supervision of Dr. Swagatam Das. I have already been introduced to clustering algorithms, including Soft and Hard clustering, using *Bregman divergences* and *F- Divergences*, DP -means, Spectral Clustering, Sparse Subspace clustering, etc. I am also introduced to many other concepts of Machine Learning which I learnt during my research period.

Academic Projects/ Short research experiences

Spring 2022: Worked with image dataset as a part of our *Statistics Comprehensive* course under the supervision of Prof. Saurabh Ghosh and Prof. Bimal Kumar Roy, where we trained a novel model to predict genre/style of a painting **without** using deep learning method to learn the features of the painting and instead defined our own features and incorporated a clustering framework to it. We have also developed few methods to analyze those paintings which do not fall into any genre/styles that were chosen to train the model.

Spring 2022: Worked in a theoretical problem titled, *Triangle-free graphs have large independence numbers* as a part of the *Random Graph* course work under the supervision of Prof. Mridul Nandi

Fall 2021: Worked with real-life *Consumer Expenditure Survey* dataset provided by NSS (National Service Scheme) where we have done equivalent adjustments to the important variable of interest to understand the trend of consumptions in different realms like Rural/Urban, Male/Female, age etc.

Fall 2021: Worked with the real-life *Time Use Survey* dataset provided by NSS (National Service Scheme) where we have estimated various key measurements (which includes different realms such as Rural/Urban, Male/Female etc.) important for a policymaker to make decisions.

Spring 2021: Did a project on Time Series data, where I worked with real-life time series data, and analysed it, under the supervision of Prof. Smarajit Bose, as a part of course work of *Statistical Methods IV*.

Fall 2020: Performed data analyses on *Fisher's Iris Data* as a part of course work of *Statistical Methods III* taught by Prof. Probal Chaudhuri.

Summer 2020: Completed a group project on the proof of *Weyl's Equidistribution Theorem* as a part of semestral assessment of *Analysis II* taught by Prof. Pradipta Bandyopadhyay

June 2020: Did an independent study on **Backtracking Algorithm** and made a **Sudoku Solver**.

Fall 2019: Performed data analyses on *Galton's Height Data* as a part of course work of *Statistical Methods I* taught by Prof. Probal Chaudhuri.

2017: Made an elementary bot using Python which played the **Google Dino Game**.

2016: Did an indepedent study, found the result on my own, on how to find the *n-th root of any real number using long division method*.

Publications/Pre-prints

(⋆ represent joint authorship)

- **Chakraborty, C.** and Ma, C., "The Statistical Benefits of Multiple Responses for Learning From Demonstrations", Preprint, submitted to ICLR 2027

- **Chakraborty, C.***, Lee, J.* and Ma, C., "Generative Collapse Does Not Imply Discriminative Collapse", submitted to ICLR 2027
- **Chakraborty, C.***, Paul., S*, Chakraborty, S. and Das. S, "Automatic Convex Clustering of High-dimensional Data with Ordered Weighted ℓ_1 Regularization", (manuscript is ready)
- **Chakraborty, C.***, Paul., S*, Chakraborty, S. and Das. S, "Clustering RNA Single Cell Sequences: A Lasso-weighted k-means approach with Ordered Weighted ℓ_1 Regularization", (manuscript is ready)
- Burden, J., **Chakraborty, C.**, Fang, Q., Lin, L., Malibari, N., Matoush, S., Milbank, I., Parekh, Z., Prado, M., Ren, R., Rong, Q., Sánchez Garza, M., Sun, E., Trevino, E., Yamada, D., "Selecting Balls From Urns With Partial Replacement Rules", published in the PUMP Journal of Undergraduate Research 2024 (Names of the authors are given alphabetically)
- **Chakraborty, C.***, Paul., S*, Chakraborty, S. and Das. S, "Clustering High-dimensional Data with Ordered Weighted ℓ_1 Regularization", published in The 26th International Conference on Artificial Intelligence and Statistics (AISTATS) 2023
- Basak,G.K., **Chakraborty, C.** and Das, P.K., (2023) "In search of an optimal public policy in a pandemic: The question of lives versus of livelihood", published in Journal of Economics Analysis (JEA)

Works in Progress

- Working with Prof. Cong Ma on a RLHF problem combining with multi-turn RL apparatus.
- Working with Prof. Frederic Koehler on a Gaussian Graphical Model problem improving the present algorithm to run in polynomial time with a better sample complexity.

Talks

- Gave a talk in the University of Pennsylvania on my work with Prof. Larry Wasserman and Prof. Arun Kumar Kuchibhotla.

Other Projects

- Android Development:
  Made few elementary apps for Android.
- Web Scrapping:
  Did many web scrapping projects using `Python`
  Made a CLI program *YouScrap*, which is used to retrieve information of any YouTube video/channel.
- Did few projects using *Arduino*.
- Cyber Security:
  Did a project on cyber security as a part of course work of **Cyber Security and Ethical Hacking using Python**.

Courses, Camps and Workshops

Fall 2020 Completed a *60 hours* course on **Cyber Security and Ethical Hacking using Python** at *Ardent Collaborations*.

October 2019 Attended a workshop on *A.I. and Machine Learning* organised jointly by **ISI-INTEL**.

Winter 2016 **INMO 2016 training camp** at *North-Eastern Hill University, Shillong*

Winter 2015 **Expedition**, a mathematics camp funded by *National Board for Higher Mathematics (NBHM)* at *G.C. College, Silchar*.

Teaching Experiences

- Currently teaching *Linear Algebra* to few students pursuing their bachelor course.

- I had a three months experience as a teacher at PiParadox where I taught programming languages like `R` and `Python`
- Conducted a three days *workshop* on `Python`, through YouTube live streaming.

Skills

Programming	Proficient in R, C, C++, Python2.7, Python3+, Shell language, Java, SQL, PHP (basics), Kotlin (basics)
Markdown /Web	LaTeX(proficient), HTML (proficient), CSS (basics)
Operating System	Proficient in both Linux and Windows. Basics in Mac OS.
Image editing	Amateur in GIMP.

Other Informations

Languages

- Bengali (Mother tongue)
- English (Second language)
- Hindi
- Oriya
- Assamese

Hobbies and Interests

- Competitive programming and coding.
- Love to learn new languages, currently learning *Sanskrit*.
- Chess (I am a **FIDE** rated chess player).
- Love classical musics and musical instrument. I play the *Esraj* and the *Bansuri* myself.
- Exploring Linux and Distro hopping.
- Love to spend time in library and read books.
- Bridge (I also took part in local Bridge tournaments).